



FHIR-Native Maternal-Child Care Continuity Agent

Agents Assemble Hackathon · Judge Packet · May 2026 · Michal Kurc (Poland)

One-sentence pitch

MamaGuard recalls prior clinician decisions from FHIR, synthesizes compound maternal + SDOH risk that rules cannot, and writes back a Goal + CarePlan + CommunicationRequest *only after the clinician approves*.

Three headline numbers

93.1%	Tier-2b end-to-end pass rate (45/47) on the April 17, 2026 submission run — live HAPI FHIR, not mocks.
+30%	Average lift over a naive LLM on the AI-factor benchmark, judged by DeepSeek v3.2 across five clinical dimensions.
100%	Clinician-review flagging on URGENT / HIGH cases in safety verification — nothing dangerous auto-commits.

What's in this packet

1. Why MamaGuard — judges' criteria mapping
2. AI Factor — naive LLM vs MamaGuard, scored
3. Safety & impact
4. Architecture & stack

1 · Why MamaGuard

A judge-facing summary mapping MamaGuard to the three Agents Assemble criteria.

AI Factor — GenAI beyond what rules can do

A naive LLM with the same FHIR data scores **52%** on clinical rubrics. MamaGuard scores **82%** — a **+30%** lift validated by DeepSeek v3.2 across three severity tiers.

- **Cross-resource synthesis** — A rule engine flags BP >140/90. MamaGuard reads 6 pregnancies with 5 losses, concurrent diabetes, postpartum timing, and housing instability — and explains why the combination is dangerous, citing 7+ FHIR references.
- **SDOH-clinical integration** — Connects a Medicaid gap + language barrier + food insecurity + postpartum BP crisis into a unified outreach plan with concrete community resources (211 / WIC / SNAP) and trackable FHIR CarePlan + Goal pairs.
- **Compound risk elevation** — A rule engine produces 5 flat flags; MamaGuard synthesizes URGENT compound risk across BP + HbA1c + pregnancy loss + housing + Medicaid gap with 5 cross-factor interactions a rule engine cannot produce.

Potential Impact — clear hypothesis for outcomes

Metric	Current baseline	Target with MamaGuard
Postpartum follow-up completion	60% attend	85%+ via proactive gap detection
Hypertensive crisis detection	Median 5 days	Same-day via BP trend analysis
Clinician chart-review time	15–20 min	<2 min AI-synthesized summary
SDOH screening completion	<25% eligible	80%+ via auto Z-code + Coverage analysis

Cost context: Preventable maternal morbidity costs \$32.3B/year in the US; each avoided severe maternal-morbidity event saves ~\$115K in acute-care costs.

Feasibility — realistic for real healthcare

- **Approval-gated write-back.** Nothing writes to the chart without clinician sign-off — the single architectural choice that answers most safety, liability, and regulatory questions.
- **Standards-native.** FHIR R4 + MCP + A2A v1 + SHARP extension spec. No proprietary lock-in.
- **Live deployment.** Running on Azure Container Apps against a live HAPI FHIR R4 server; not a screenshot demo.

2 · AI Factor — Scored Comparison

DeepSeek v3.2 was used as an independent LLM-as-judge across three patient severity tiers. Both systems were given identical FHIR data; only the agent architecture differed.

Dimension	Naive LLM	MamaGuard	Lift
Clinical accuracy	65%	85%	+20%
Risk assessment	47%	82%	+35%
Safety	38%	80%	+42%
Completeness	45%	80%	+35%
Output quality	65%	83%	+18%
Overall	52%	82%	+30%

What the lift comes from

- FHIR-native memory.** MamaGuard recalls a prior `DocumentReference` note ("Dr. Kim, 2026-04-02 — metformin declined for GI intolerance") and refuses to re-recommend it. The naive LLM has no continuity across sessions and re-suggests metformin every time.
- Compound reasoning over isolated flags.** The naive LLM emits five disconnected bullets ("BP elevated", "HbA1c high", "coverage gap"). MamaGuard produces a single clinical narrative: *"Stage 2 postpartum HTN + diabetes-range HbA1c + Medicaid gap + Spanish-only support — call her today."*
- Plan-mode discipline.** The naive LLM either refuses to act or hallucinates writes. MamaGuard generates a structured pending bundle (Goal + CarePlan + CommunicationRequest) that the clinician can approve or reject as one atomic decision.

Evidence: `benchmarks/fixtures/ai_factor_comparison.{md,json}`, run 2026-04-17.

3 · Safety & Impact

Safety verification — 100% URGENT / HIGH flagging

Severity tier	Cases	Flagged for clinician review	Auto-committed
URGENT (postpartum HTN crisis, DKA risk)	12	12 / 12 (100%)	0
HIGH (compound maternal + SDOH)	18	18 / 18 (100%)	0
MEDIUM (single-factor maternal risk)	11	11 / 11 flagged for review	0
LOW (routine follow-up)	6	6 / 6 flagged for review	0

Every clinical action — without exception — surfaces as a pending FHIR bundle awaiting clinician approval. This is the architectural answer to the safety, liability, and regulatory questions every MD judge will ask.

Potential impact — populations and dollars

- **Target cohort.** Perinatal Medicaid populations — where compound maternal + SDOH risk hits hardest and continuity gaps are largest.
- **US maternal mortality.** Worst in the developed world; ~80% of deaths are preventable; Black women face 3–4× higher mortality.
- **Cost wedge.** \$32.3B/year in preventable maternal morbidity; ~\$115K saved per avoided severe-morbidity event.
- **Time wedge.** 15–20 min of chart review compressed to <2 min of AI-synthesized risk narrative, with provenance citations to FHIR resources.

What this isn't

MamaGuard is not a diagnostic tool, not a triage replacement, and not autonomous. It is a continuity, synthesis, and outreach-planning layer that runs *under* clinician approval. The clinician is always the decision-maker.

4 · Architecture & Stack

MamaGuard is an orchestrator + three specialist sub-agents over a shared FHIR continuity layer, exposed through MCP for tool reuse and A2A v1 for conversational invocation from Prompt Opinion.



Layer	Choice
Agent runtime	Google ADK (Python) · orchestrator + Maternal / Pediatric / SDOH specialists
Production model	Gemma-4-26B-A4B (FP8) — SFT-fine-tuned on Nemotron-generated gold responses
Benchmark / gold-generation model	Nemotron-3-Super-120B (NVFP4, 262K ctx) on DGX Spark via 10GbE
LLM-as-judge	DeepSeek v3.2 (OpenRouter)
Interop standards	FHIR R4 · MCP · A2A v1 · SHARP extension spec · SMART-on-FHIR scopes
Deployment	Azure Container Apps · HAPI FHIR R4 (H2 in-memory)
Inference serving	vLLM (Nemotron and Gemma 4, FP8 / NVFP4)

Model pipeline (Nemotron → Gemma 4)

1. Nemotron-3-Super-120B generates gold-quality responses across the maternal-child benchmark suite.
2. Gemma-4-26B-A4B is SFT-fine-tuned on those gold responses (small enough to train on a single DGX Spark, 4B active params for fast inference).
3. DeepSeek v3.2 judges both — Gemma 4 retains the +30% AI-factor lift while serving at clinic-budget cost.